

Unit 1 Assignment: The Model Benchmark Challenge

NAME : NISHA SURESH PATIL

SRN : PES2UG23CS393

SECTION: F

Deliverable: Observation Table

Task	Model	Classification (Success/Failure)	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
Generation	BERT	Failure	Generated only dots, no meaningful continuation.	BERT is an Encoder-only model and lacks autoregressive next-token prediction.
Generation	RoBERTa	Failure	Did not generate any continuation.	RoBERTa is also Encoder-only and not trained for text generation.
Generation	BART	Failure	Generated random and incoherent words.	BART was loaded as causal LM without fine-tuning, leading to unstable generation.
Fill-Mask	BERT	Success	Correctly predicted words like "create", "generate", "produce".	BERT is trained using Masked Language Modeling (MLM).
Fill-Mask	RoBERTa	Failure	Failed due to missing <mask> token.	RoBERTa uses <mask> token instead of [MASK].
Fill-Mask	BART	Failure	Failed due to missing <mask> token.	BART also uses <mask> token, not [MASK].
QA	BERT	Partial Success	Returned incomplete answer: ", and deepfakes".	Base BERT is not fine-tuned for QA.
QA	RoBERTa	Success(for this run)	Returned full correct answer.(can vary for different runs)	Even without fine-tuning, encoder models can extract spans.
QA	BART	Partial Success	Returned partial phrase: "significant risks such as".	BART not fine-tuned for extractive QA.